

# Emulator

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## Setup

1. Clone the current version of the emulator from the github repository
2. Install Rust using the appropriate installation method for your operating system found at <https://www.rust-lang.org/tools/install>
3. From within your terminal find the emulator directory and run "cargo build". If the build finishes then you've successfully installed the Rust toolchain and are ready to use the emulator.

## Usage

**Function:** "cargo run" is used to run programs in Rust.

**Parameters:** The emulator takes two arguments as function parameters. The first is the path to a .mem file for execution. There is an example file located in "emulator/output.mem". The second parameter is optional and describes the mode of execution for the emulator. The mode using a terminal UI requires the parameter "-u". If you wish to run the emulator in headless mode, use the "-h" argument. The third execution mode "-b" is a headless mode that allows you to define a third argument specifying the number of cycles the emulator should run before stopping.

**UI Mode:**

```
cargo run output.mem -u
```

**Headless Mode:**

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```
cargo run output.mem -h
```

### Breakpoint Mode:

```
cargo run output.mem -b 10
```

**Usage:** When using UI mode, press “f” to toggle freerunning. Use “Enter” to step through the program. Use “q” to quit and dump memory. While running in UI mode pressing “h”, “s”, or “u” will toggle the register display between hex, signed, or unsigned accordingly.

**Output:** The emulator produces two output files: “dataoutput.csv” and “memoutput.csv”. “dataoutput.csv” contains value and address pairs where the value is not zero. “memoutput.csv” contains the entire address space.

**Configuration:** The emulator uses a config.json file found in src/ to specify the emulator configuration. The “memory” field within this file defines the size of the memory array used by the emulator. The “tickRate” field within this file changes the tick rate at which the emulator freeruns in UI mode.

**User Interface:** Within “config.json” there are fields that can be used to change the size of specific blocks and the width of columns within the UI. These can be adjusted in the event that UI elements don’t properly fit on screen.

## File Structure

**main.rs:** This file is the entry point for the program and it handles arguments, configuration, and starts the emulator in the appropriate mode.

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**emulator.rs:** This file contains the state and execution of the CPU model. New operations should be added to the match statement within this file.

**ui.rs:** This file contains the UI code called from “main.rs” when the “-u” argument is chosen. The CPU component is instantiated within this file in UI mode. The emulator in headless mode does not make use of this file.